

BioNORAD: Fast Scalable Pandemic Risk Assessment of Influenza A Strains Circulating In Non-human Hosts

Animal influenza viruses emerging into humans have triggered devastating pandemics in the past. Yet, our ability to evaluate the pandemic potential of individual strains that do not yet circulate in humans, remains limited. Here we propose to develop an experimentally validated platform called the Emergenet (Enet), to predict in near-real-time where and when new variants of concern would emerge, using only observed sequences of key viral proteins, procured in ongoing global surveillance of Influenza A viruses. We bring together new machine learning algorithms customized to the problem at hand, key insights from information theory, evolutionary theory, epidemiology and precise statistical uncertainty quantification to develop a rigorous framework, to track evolutionary trajectories of pathogens through a complex, poorly characterized, and dynamically changing fitness landscape. Our deliverable is best described as the foundations for creating a platform akin to bio-NORAD, *identifying when and where an imminent zoonotic emergence event is likely, and if such novel strains are likely to achieve human-to-human transmission capability*.

Influenza viruses constantly evolve¹, sufficiently altering surface protein structures to evade the prevailing host immunity, and cause the recurring seasonal epidemic. These periodic infection peaks claim a quarter to half a million lives² globally. Additionally, Influenza A, partly on account of its segmented genome and its wide prevalence in animal hosts, can easily incorporate genes from multiple strains and (re)emerge as novel human pathogens³, thus harboring a high pandemic potential. Strains spilling over into humans from animal reservoirs is thought to have triggered pandemics at least four times (1918 Spanish flu/H1N1, 1957 Asian flu/H2N2, 1968 Hong Kong flu/H3N2, 2009 swine flu/H1N1) in the past 100 years⁴. One approach to mitigating such risk is to identify animal strains that do not yet circulate in humans, but is likely to spill-over and quickly achieve human-to-human (HH) transmission capability. While global surveillance efforts collect wild specimens from diverse hosts and geo-locations annually, our ability to objectively, reliably and scalably risk-rank individual strains remains limited⁵. The Center for Disease Control's (CDC) current solution to this problem is the Influenza Risk Assessment Tool (IRAT)⁶, which relies on time-consuming proteomics and transmission assays and potentially subjective evaluations by subject matter experts, taking weeks to months to compile for each strain of concern. With tens of thousands of strains being sequenced annually, this results in a scalability bottleneck.

Here we plan to develop a platform powered by novel pattern discovery and recognition algorithms to automatically parse out emergent evolutionary constraints operating on Influenza A viruses in the wild, to provide a less-heuristic theory-backed scalable solution to emergence prediction. We plan to show that this capability enables preempting strains which are expected to be in future human circulation, and approximate IRAT scores of non-human strains without experimental assays or SME scoring, in second as opposed to weeks or months. Our approach automatically takes into account the time-sensitive variations in selection pressures as the background strain circulation changes over time, and will potentially be able to rank-order strains adaptively. Additionally, we plan to validate our ability to predict future variations of viral proteins by showing that predicted variants of HA and NA fold correctly, and are functional, binding to the relevant human receptors in in-vivo laboratory experiments. Thus, bringing together rigorous data-driven modeling, and validation via tools from reverse genetics we plan to deliver an actionable and deployable platform that optimally exploits the current biosurveillance capacity.

The BioNORAD platform will enable proactive and actionable global surveillance for emerging pandemic threats from Influenza A. This importance of the ability to preempt pandemic risk to the national interest of the United States cannot be overstated, especially in the context of protecting DoD assets and personnel deployed in potentially high risk centers of emergence. Additionally, the BioNORAD will enable preemptive action including the inoculation of animal reservoirs before the first human infection, potentially eliminating the pandemic before it has a chance to trigger.

Emergenet: A Digital Twin for Predicting Viral Emergence and Global Biosurveillance

Scientific Rationale and Need Influenza A viruses continue to pose a major pandemic threat due to their segmented genome and the ability to reassort and mutate within animal reservoirs. Historical influenza pandemics, including those in 1918, 1957, 1968, and 2009, originated from zoonotic spillover events, demonstrating the ongoing risk of novel strains emerging. Despite global biosurveillance efforts that collect thousands of viral sequences annually, current methods for assessing the potential spillover risk remain limited. The CDC's Influenza Risk Assessment Tool (IRAT) relies on expert-driven evaluation and time-intensive laboratory assays, which are not scalable for real-time global risk assessment.

To address this gap, Emergenet is designed as a computational framework that constructs a digital twin of viral sequence evolution, enabling real-time forecasting of potential zoonotic spillover events. This approach provides a scalable, data-driven alternative to traditional expert evaluation by rapidly assessing viral emergence risk based on mutational dependencies inferred from large-scale genomic datasets. The framework is further extended into BioNORAD, an operational prototype for continuous global biosurveillance that monitors and prioritizes high-risk viral strains before human transmission occurs.

Objectives and Aims The project will begin with the development and validation of the Emergenet platform. This will involve constructing probabilistic models that capture cross-mutation dependencies in influenza genomes using publicly available sequence databases such as NCBI and GISAID. The Emergenet system will incorporate a novel emergence-distance (E-distance) metric to quantify the probability of a given viral strain evolving into a more transmissible or virulent form. By ranking animal strains based on their pandemic potential, this system will allow for rapid triaging of surveillance efforts. Predictions generated by Emergenet will be compared with historical IRAT assessments and past

pandemic strains to evaluate its predictive accuracy. Retrospective analysis will determine whether Emergenet could have identified past pandemic strains before they achieved human-to-human transmission.

The second phase of the project involves experimental validation using reverse genetics. High-risk viral variants predicted by Emergenet will be generated in the lab through targeted mutagenesis. These engineered strains will be assessed for replication efficiency, host adaptation, and fitness in human lung epithelial cell lines and primary lung cells. Additional experiments will evaluate viral entry efficiency using receptor-binding assays, and antigenic properties will be analyzed through hemagglutination inhibition (HAI) tests. Competitive fitness assays will further determine whether Emergenet-predicted strains demonstrate increased replication efficiency compared to wild-type parental strains. The goal of these experiments is to confirm that mutations flagged by Emergenet indeed enhance viral fitness, while randomly selected mutations generally disrupt viral viability.

Clarification Regarding Gain-of-Function Research It is important to emphasize that this project does not constitute gain-of-function (GoF) research as it does not involve engineering viruses to acquire new functions that increase transmissibility or pathogenicity beyond their natural evolutionary constraints. The mutations introduced during experimental validation are strictly limited to those that have already been observed in naturally circulating strains, ensuring that no novel, unnatural, or artificially enhanced viral phenotypes are generated. Additionally, all reverse genetics experiments will be conducted under stringent biosafety regulations (BSL-2/BSL-3 as required), with oversight from institutional biosafety committees. The primary goal of these experiments is to validate the accuracy of Emergenet's predictive modeling in identifying naturally emerging viral variants, rather than generating more virulent or transmissible viruses. The focus remains on understanding evolutionary trajectories rather than enhancing viral capabilities, aligning fully with ethical and biosafety considerations.

The third phase of the project will deploy BioNORAD, a real-time biosurveillance system integrating Emergenet's predictive capabilities. BioNORAD will automatically process and score new influenza sequences from global surveillance databases, continuously updating risk profiles for emerging strains. A user-friendly visualization dashboard will allow public health officials to track high-risk strains in real time, ensuring that actionable intelligence is available for pandemic preparedness. The system's predictions will be evaluated by monitoring whether strains identified as high-risk later demonstrate increased human transmissibility or lead to outbreak events.

Anticipated Outcomes and Impact The Emergenet framework will revolutionize pandemic risk assessment by enabling a scalable, AI-driven alternative to traditional expert evaluations. By providing real-time forecasting of viral emergence, the system will significantly enhance global preparedness efforts. The integration of BioNORAD will allow for preemptive action before zoonotic spillovers occur, ensuring that resources are allocated efficiently to monitor and contain high-risk viral strains. Experimental validation of Emergenet's predictions will provide critical empirical support, demonstrating that the computational model can reliably identify biologically viable emergent strains. Ultimately, this project establishes a novel paradigm for pandemic prevention, transforming biosurveillance from a reactive system into a proactive global defense strategy.

PIs for the proposed effort are Ishanu Chattopadhyay (lead, machine learning, UChicago) and Balaji Manicassamy (virology, Iowa State), and will be conducted over 2 years with a direct cost of 200K USD.

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